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A theoretical model for the stick/bounce behaviour of adhesive, elastic–plastic spheres

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Abstract

The paper considers the normal impact of elastic–perfectly plastic spheres, with and without interface adhesion, and presents an analytical solution for the coefficient of restitution which is expressed in terms of the impact velocity, the critical sticking velocity and the velocity below which the interaction is assumed to be elastic.

Keywords: Impact; Restitution; Adhesion; Contact mechanics

1. Introduction

Numerical simulations of gas-particle flows are now commonly applied to many industrial problems. However, whether the simulations track individual particles or consider the particle phase to be a continuum, the way in which inter-particle and particle-wall collisions are modelled has a significant influence on the flow predictions. The central issue which has to be addressed is whether particles stick or rebound upon collision and, in the case of rebound, the changes that occur in both the linear and rotational kinetic energy. Nevertheless, many researchers assume a constant coefficient of restitution and, in areas such as filtration and particle deposition studies, it is often assumed that particles which come into contact with each other or with a containing surface do not rebound.

The objective of the work presented here is to obtain an analytical solution for the coefficient of restitution based on the concepts of theoretical contact mechanics. We first of all consider the case of elastic–perfectly plastic spheres with no interface energy and obtain a velocity dependent coefficient of restitution which, at high impact velocities, is inversely proportional to the velocity raised to the power 0.25. We then examine the case of elastic–adhesive spheres and derive equations for the critical sticking velocity below which rebound does not occur and the coefficient of restitution for velocities above the sticking velocity. Finally, by assuming that the work dissipated by plastic work and the work dissipated due to interface adhesion are additive, we obtain a general solution for the coefficient of restitution for adhesive, elastic–plastic spheres.

2. Elastic–perfectly plastic spheres

For two contacting spheres of radii R_i and elastic properties E_i and ν_i ($i=1,2$) the Hertzian pressure distribution over the contact area of radius a is

$$p(r)=\frac{3P}{2\pi a^2}\left[1-\left(\frac{r}{a}\right)^2\right]^{1/2} \tag{1}$$

The contact normal force P and contact radius a are given by

$$P=\frac{4}{3}E^*R^{*1/2}\alpha^{3/2} \tag{2}$$

$$a=\left(\frac{3PR^*}{4E^*}\right)^{1/3} \tag{3}$$

where α is the relative approach of the two particle centroids and

$$a^2=R^*\alpha \tag{4}$$

In the above equations

$$\frac{1}{E^*}=\frac{1-\nu_1^2}{E_1}+\frac{1-\nu_2^2}{E_2} \tag{5}$$

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$$\frac{1}{R^*}=\frac{1}{R_1}+\frac{1}{R_2} \tag{6}$$

If the relative impact velocity V is just large enough to initiate yield in one of the spheres then, using Eqs. (2) and (4) we may write

$$\frac{1}{2}m^*V_y^2=\int_0^{\alpha_y}Pd\alpha=\frac{8E^*a_y^5}{15R^{*2}} \tag{7}$$

where V_y, which we define as the yield velocity, is the relative impact velocity below which the interaction behaviour is assumed to be elastic, a_y is the contact radius when yield occurs and m^* is related to the particle masses m_i by the equation

$$\frac{1}{m^*}=\frac{1}{m_1}+\frac{1}{m_2} \tag{8}$$

We now define a 'limiting contact pressure' $p_y=p_0$ when $a=a_y$. Using Eqs. (1) and (3),

$$p_y=\frac{2E^*a_y}{\pi R^*} \tag{9}$$

Combining Eqs. (7) and (9), we obtain

$$V_y=\left(\frac{\pi}{2E^*}\right)^2\left(\frac{8\pi R^{*3}}{15m^*}\right)^{1/2}p_y^{5/2}=3.194\left(\frac{p_y^5R^{*3}}{E^{*4}m^*}\right)^{1/2} \tag{10}$$

In the case of a sphere of density ρ impacting with a plane surface, $R^*=R$, $m^*=m$ and Eq. (10) reduces to

$$V_y=\left(\frac{\pi}{2E^*}\right)^2\left(\frac{2}{5\rho}\right)^{1/2}p_y^{5/2}=1.56\left(\frac{p_y^5}{E^{*4}\rho}\right)^{1/2} \tag{11}$$

which was originally obtained by Davies [1].

In order to model the post-'yield' behaviour, it is necessary to make some simplifying assumptions. If plastic deformation occurs we assume a Hertzian pressure distribution with a cut-off corresponding to the limiting contact pressure p_y defined by Eq. (9). After yield, the normal force is given by

$$P=P_e-2\pi\int_0^{a_p}[p(r)-p_y]r\mathrm{d}r \tag{12}$$

where P_e is the equivalent elastic force given by Eq. (3) which would result in the same total contact area and a_p is the radius of the contact area over which a uniform pressure is assumed, as indicated in Fig. 1. Integrating Eq. (12) we obtain

$$P=\pi a_p^2p_y+P_e\left[1-\left(\frac{a_p}{a}\right)^2\right]^{3/2} \tag{13}$$

Considering the conditions at yield, p_y may be defined in terms of the normal force P_y and contact radius a_y as

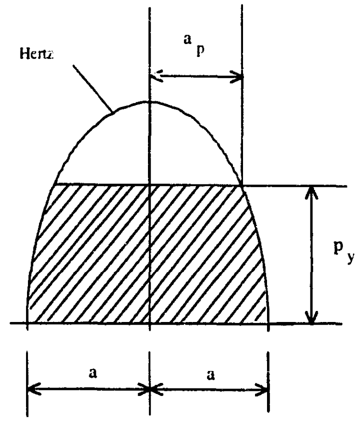

Fig. 1. Normal traction distribution for elastic–perfectly plastic spheres.

$$p_y=\frac{3P_y}{2\pi a_y^2} \tag{14}$$

or, according to Fig. 1,

$$p_y=\frac{3P_e}{2\pi a^2}\left[1-\left(\frac{a_p}{a}\right)^2\right]^{1/2} \tag{15}$$

The contact radius a is obtained from

$$a^3=\frac{3R^*P_e}{4E^*} \tag{16}$$

Hence, combining Eqs. (14)–(16) we find that

$$\left[1-\left(\frac{a_p}{a}\right)^2\right]=\left(\frac{a_y}{a}\right)^2 \text{ or } a^2=a_p^2+a_y^2 \tag{17}$$

Substituting Eqs. (16) and (17) into Eq. (13) we obtain

$$P=P_y+\pi p_y(a^2-a_y^2) \tag{18}$$

Substituting Eq. (4), the force–displacement relationship during plastic loading is given as

$$P=P_y+\pi p_yR^*(\alpha-\alpha_y) \tag{19}$$

which is linear, as shown in Fig. 2.

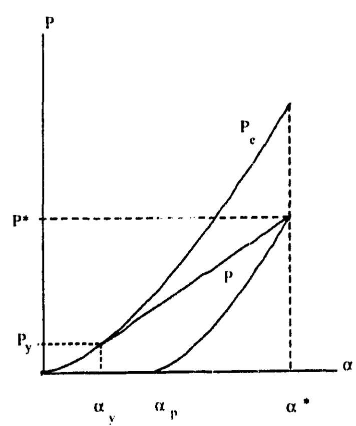

Fig. 2. Force–displacement relationship for elastic–perfectly plastic spheres.

If plastic deformation occurs during the loading stage the contact curvature during unloading is $1/R_p^* < 1/R^*$ due to permanent deformation of the contact surfaces. During unloading the force–displacement behaviour is assumed to be elastic and is provided by the Hertzian equations but with a curvature $1/R_p^*$ corresponding to the point of maximum compression. At the point of unloading, the contact area developed by the actual maximum normal force P^* and reduced curvature $1/R_p^*$ is the same as that which would be generated by an equivalent elastic force P_e^* and a contact curvature $1/R^*$. Hence, from Eq. (16),

$$R_p^* P^* = R^* P_e^* \tag{20}$$

where

$$P_e^* = \frac{4}{3} E^* R^{*1/2} \alpha^{*3/2} \tag{21}$$

It can be seen from Fig. 2 that the linear plastic loading curve is tangential to the Hertzian curve at the yield point and, when extended, intersects the vertical axis at $P_0 < 0$. From Eqs. (18) and (9) the plastic contact stiffness is defined as

$$k_N = \pi R^* p_y = 2E^* a_y \tag{22}$$

Therefore,

$$\alpha^* = \frac{P^* - P_0}{\pi R^* p_y} \tag{23}$$

and

$$P_0 = P_y - 2E^* a_y \alpha_y \tag{24}$$

Substituting Eqs. (3) and (4),

$$P_0 = -\frac{P_y}{2} \tag{25}$$

and, hence,

$$\alpha^* = \frac{2P^* + P_y}{2\pi R^* p_y} \tag{26}$$

$$\alpha^{*2} = \frac{2P^* + P_y}{2\pi^* p_y} \tag{27}$$

Combining (20), (21) and (26) leads to

$$R_p^* = \frac{4E^*}{3P^*}\left(\frac{2P^* + P_y}{2\pi p_y}\right)^{3/2} \tag{28}$$

and, during elastic unloading,

$$P = \frac{4}{3} E^* R_p^{*1/2} (\alpha - \alpha_p)^{3/2} \tag{29}$$

where α_p is defined in Fig. 2. Numerical simulations, in which the contact interactions were based on the above theoretical considerations, have been performed and Fig. 3 illustrates the results obtained for different impact velocities. The figure shows how the normal contact force–displacement response

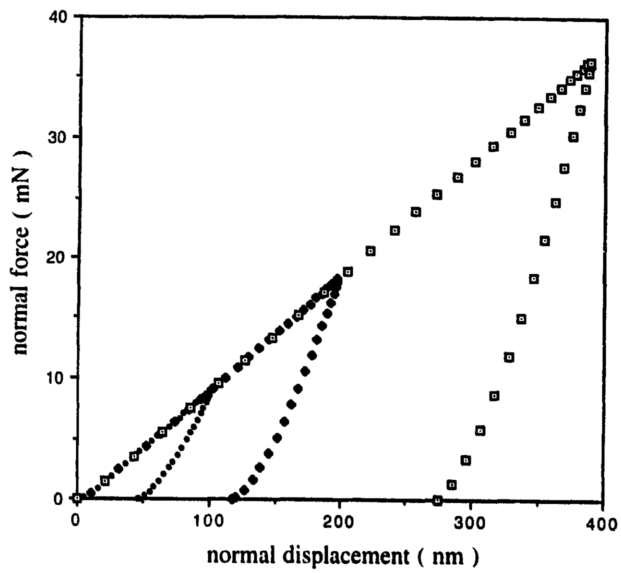

Fig. 3. Effect of impact velocity on the normal contact force–displacement response for elastic–perfectly plastic spheres.

is affected by the severity of the impact. When the impact velocity is greater than the yield velocity, the unloading stiffness increases with increase in impact velocity. Consequently, the ratio of work done during elastic recovery to the work done during compression depends on the magnitude of the impact velocity and therefore the coefficient of restitution is velocity dependent, as demonstrated in the next section.

3. Coefficient of restitution

At impact velocities $V_i \leq V_y$ no plastic deformation occurs and, ignoring energy losses due to elastic wave motion in the two bodies, the coefficient of restitution $e = 1.0$. If the impact velocity $V_i > V_y$, the rebound kinetic energy is equal to the work done during elastic recovery. Thus,

$$\frac{1}{2} m^* V_r^2 = \frac{2}{5} P^* (\alpha^* - \alpha_p) \tag{30}$$

where

$$(\alpha^* - \alpha_p) = \frac{a^{*2}}{R_p^*} \tag{31}$$

Hence, using Eqs. (27) and (28)

$$\frac{1}{2} m^* V_r^2 = \frac{3P^{*2}}{10E^* a^*} \tag{32}$$

The coefficient of restitution is defined as $e = V_r / V_i$. Thus.

$$e^2 = \frac{3P^{*2}}{5E^* a^* m^* V_i^2} \tag{33}$$

The initial kinetic energy is equal to the work done during decelleration of the particles. Therefore, from inspection of Fig. 2,

$$\frac{1}{2} m^* V_i^2 = \frac{2}{5} P_y \alpha_y + \frac{1}{2} (P_y + P^*)(\alpha^* - \alpha_y) \tag{34}$$

Using Eqs. (4), (14) and (26)

$$\frac{1}{2}m^*V_i^2=\frac{2}{5}P_y\alpha_y+\frac{1}{2}(P_y+P^*)(\alpha^*-\alpha_y) \tag{35}$$

Therefore,

$$\frac{1}{2}m^*V_i^2=\frac{2}{5}P_y\alpha_y+\frac{P^{*2}-P_y^2}{2\pi R^*p_y} \tag{36}$$

Using Eqs. (2) and (22) we obtain

$$\frac{1}{2}m^*V_i^2=\frac{2}{5}P_y\alpha_y+\frac{P^{*2}-P_y^2}{2\pi R^*p_y} \tag{37}$$

From which

$$P^{*2}=2E^*a_y\left(m^*V_i^2-\frac{1}{6}m^*V_y^2\right) \tag{38}$$

Substituting Eq. (38) into Eq. (33) the coefficient of restitution can be obtained from

$$e^2=\frac{6a_y}{5a^*}\left[1-\frac{1}{6}\left(\frac{V_y}{V_i}\right)^2\right] \tag{39}$$

Using the substitutions

$$a_y^2=\frac{3P_y}{2\pi p_y} \tag{40}$$

and

$$a^{*2}=\frac{2P^*+P_y}{2\pi p_y} \tag{41}$$

we obtain

$$e^2=\frac{6\sqrt{3}}{5}\left(\frac{P_y}{2P^*+P_y}\right)^{1/2}\left[1-\frac{1}{6}\left(\frac{V_y}{V_i}\right)^2\right] \tag{42}$$

Combining Eqs. (2), (7) and (38)

$$\frac{P^*}{P_y}=\left[\frac{6}{5}\left(\frac{V_i}{V_y}\right)^2-\frac{1}{5}\right]^{1/2} \tag{43}$$

Therefore, the general expression for the coefficient of restitution is

$$e=\left(\frac{6\sqrt{3}}{5}\right)^{1/2}\left[1-\frac{1}{6}\left(\frac{V_y}{V_i}\right)^2\right]^{1/2}\times\left[\frac{\left(\frac{V_y}{V_i}\right)}{\left(\frac{V_y}{V_i}\right)+2\sqrt{\frac{6}{5}-\frac{1}{5}\left(\frac{V_y}{V_i}\right)^2}}\right]^{1/4} \tag{44}$$

with the yield velocity V_y defined by Eq. (10) or, in the special case of a sphere impacting a plane surface, by Eq. (12). Eq. (44) satisfies the condition $e=1.0$ when $V_i=V_y$. At high velocities, $(V_y/V_i)^2\to 0$ and Eq. (44) becomes

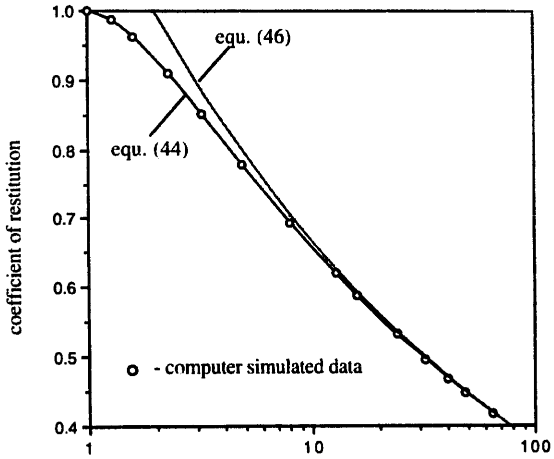

Fig. 4. Coefficient of restitution for elastic–perfectly plastic spheres.

$$e=\left(\frac{6\sqrt{3}}{5}\right)^{1/2}\left[\frac{V_y}{V_y+\frac{2\sqrt{6}}{\sqrt{5}}V_i}\right]^{1/4} \tag{45}$$

Taking $V_i\gg V_y$ we then obtain

$$e=\left(\frac{6\sqrt{3}}{5}\right)^{1/2}\left(\frac{\sqrt{5}}{2\sqrt{6}}\right)^{1/4}\left(\frac{V_y}{V_i}\right)^{1/4}=1.185\left(\frac{V_y}{V_i}\right)^{1/4} \tag{46}$$

For the case of a sphere impacting a plane surface we may substitute in Eq. (46) for V_y using Eq. (11) to obtain

$$e=1.324\left(\frac{p_y^5}{E^{*4}\rho}\right)^{1/8}(V_i)^{-1/4} \tag{47}$$

which was given by Thornton and Ning [2]. Johnson [3] provided a similar expression to Eq. (47) except that the prefactor was 1.72 as a result of assuming that the plastic normal contact stiffness was twice that given by Eq. (22).

Fig. 4 shows the dependence of the coefficient of restitution on the impact velocity according to the general expression (44). Superimposed on the figure is the prediction according to Eq. (46) and the numerical results obtained by Thornton and Ning [2]. It can be seen that Eq. (46) is only satisfactory for $V>10V_y$.

4. Adhesive elastic spheres

For adhesive elastic spheres the theory of Johnson et al. [4], commonly known as JKR theory, is assumed. Johnson [5] provided the following relationship between the contact force and relative approach

$$\frac{\alpha}{\alpha_f}=\frac{3\left(\frac{P}{P_c}\right)+2\pm 2\left(1+\frac{P}{P_c}\right)^{1/2}}{3^{2/3}\left[\frac{P}{P_c}+2\pm 2\left(1+\frac{P}{P_c}\right)^{1/2}\right]^{1/3}} \tag{48}$$

with

$$\alpha_f=\frac{3}{4}\left(\frac{\pi^2\Gamma^2R^*}{E^{*2}}\right)^{1/3}=\left(\frac{3P_c^2}{16R^*E^{*2}}\right)^{1/3} \tag{49}$$

where Γ is the interface energy and

$$P_c=\frac{3}{2}\pi\Gamma R^* \tag{50}$$

Fig. 5 shows the force displacement relationship in dimensionless form as defined by Eq. (48).

The following explanation of the stick/bounce criterion was provided by Professor K.L. Johnson in an oral contribution to a meeting of the Institute of Physics (Tribology Group) on 'Adhesive Forces in Powder Flows' at the University of Surrey in September 1985.

According to JKR theory, when two colliding surfaces come into contact the normal force between the two bodies will immediately drop to a certain value, $P=-8P_c/9$ (point A in Fig. 5), due to Van der Waals attractive forces. The velocity of the sphere is then reduced gradually and part of the initial kinetic energy is radiated into the substrate as elastic waves. When the contact force reaches a maximum value (say point B in Fig. 5) the particle velocity has been reduced to zero and the incoming stage is complete. In the recovery stage the stored elastic energy is released and converted into kinetic energy and the particle moves in the opposite direction. All the work done during the loading stage has been recovered when point A is reached during the recovery stage. However, at this point, when $\alpha=0$, the sphere remains adhered to the target and further work is required to separate the surfaces. As shown in Fig. 5, separation occurs at point F and hence the work required to break the contact W_s is given by the area under the curve for $-1<\alpha/\alpha_f<0$. Johnson suggested that a sufficiently accurate expression for W_s was

$$W_s=P_c\alpha_f=7.58\left(\frac{\Gamma^5R^{*4}}{E^{*2}}\right)^{1/3} \tag{51}$$

However, Eq. (51) may be integrated between the limits 0 and $-\alpha_f$ to obtain

$$W_s=\int_0^{\alpha_f}P\,d\alpha=0.9355P_c\alpha_f=7.09\left(\frac{\Gamma^5R^{*4}}{E^{*2}}\right)^{1/3} \tag{52}$$

which was also obtained, in a different manner, by Johnson and Pollock [6].

Neglecting energy losses due to elastic wave propagation, the only work dissipated during a collision is the work done in separating the surfaces W_s. Therefore, we may write

$$\frac{1}{2}m^*V_i^2-\frac{1}{2}m^*V_r^2=W_s \tag{53}$$

If the rebound velocity $V_r=0$ then the impact velocity $V_i=V_s$ the critical velocity below which sticking occurs and from Eqs. (52) and (53) we obtain the sticking criterion

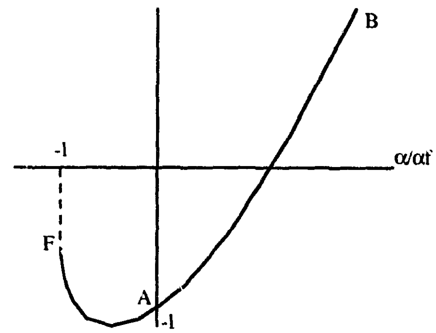

Fig. 5. Force–displacement relationship for adhesive elastic spheres.

$$V_s=\left(\frac{14.18}{m^*}\right)^{1/2}\left(\frac{\Gamma^5R^{*4}}{E^{*2}}\right)^{1/6} \tag{54}$$

For a sphere impacting a flat surface, $R^*=R$ and $m^*=m$, leading to

$$V_s=1.84\left[\frac{(\Gamma/R)^5}{\rho^3E^{*2}}\right]^{1/6} \tag{55}$$

If $V_i>V_s$ then bounce occurs and we may rewrite Eq. (53) as

$$1-\left(\frac{V_r}{V_i}\right)^2=\left(\frac{V_s}{V_i}\right)^2 \tag{56}$$

which is equivalent to Eq. (14) in Ref. [6] and from which we define the coefficient of restitution by

$$e=\left[1-\left(\frac{V_s}{V_i}\right)^2\right]^{1/2} \tag{57}$$

Fig. 6 illustrates how the coefficient of restitution varies with impact velocity. When the impact velocity is ten times higher than the critical sticking velocity the coefficient of restitution is 0.995.

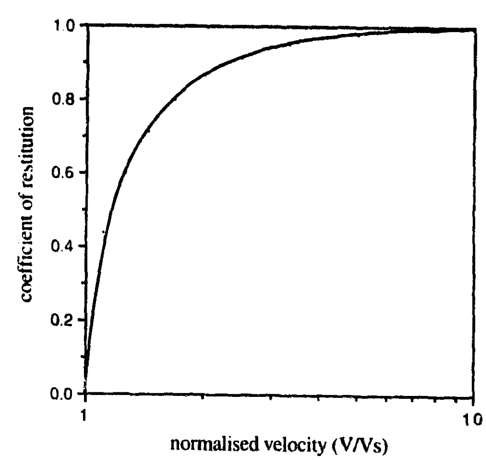

Fig. 6. Coefficient of restitution for adhesive elastic spheres.

5. Elastic–perfectly plastic adhesive spheres

Elastic–plastic impact of adhesive spheres was studied by Rogers and Reed [7] who neglected any effects of adhesion during the loading stage. They also assumed that the deformation due to impact takes the form of a central area of plastic deformation surrounded by an annulus of elastic deformation. This assumption, however, is not supported by the results of finite element analyses, e.g., Hardy et al. [8]. Hardy et al. [8] showed that the contact pressure distribution changes from the elastic elliptical shape to an essentially uniform pressure distribution as the load increases. Plastic deformation was initiated below the contact surface in the centre of the contact area and the plastic deformation zone spread until, when the contact force was ca. six times the contact force at initial yield, it reached the contact surface *at the perimeter of the contact area*. With further loading the plastic deformation zone continued to enlarge but there remained an elastic region at the surface in the centre of the contact area. The persistence of an 'elastic enclave' in the centre of the contact area during elasto-plastic deformation was also observed by Sinclair et al. [9]. In the analysis presented below, unlike Rogers and Reed [7], we account for adhesion effects during the loading stage and, although we use Eq. (17) as an approximation, this does not imply a central area of plastic deformation surrounded by an annulus of elastic deformation. Eq. (17) was obtained simply from the assumed geometry of the contact pressure distribution for a non-adhesive elasto-plastic sphere.

As demonstrated by Maugis and Pollock [10], adhesion significantly affects the subsurface stress distribution leading to plastic deformation near the perimeter of the contact area. This results in a truncation of the JKR tensile contact pressure distribution over an outer annulus of the contact area and will affect the conditions for particles to stick to each other. Fichman and Pnueli [11] obtained an expression for the work dissipated due to the adhesion induced plastic deformation but the complexity precludes its use here. However, very small velocities are normally sufficient to initiate plastic yield below the centre of the contact area and when this velocity induced plastic deformation zone spreads to reach the surface at the contact perimeter its effect will dominate over the adhesion induced plastic deformation annulus which reduces in thickness with applied loading. Consequently, we neglect the adhesion induced plastic deformation at the contact perimeter in the analysis presented below.

According to JKR theory, the normal pressure distribution over the contact area, illustrated in Fig. 7, is given by

$$p(r)=\frac{2E^*a}{\pi R^*}\left[1-\left(\frac{r}{a}\right)^2\right]^{1/2}-\left(\frac{2\Gamma E^*}{\pi a}\right)^{1/2}\left[1-\left(\frac{r}{a}\right)^2\right]^{-1/2} \tag{58}$$

The radius of the area of contact is defined by

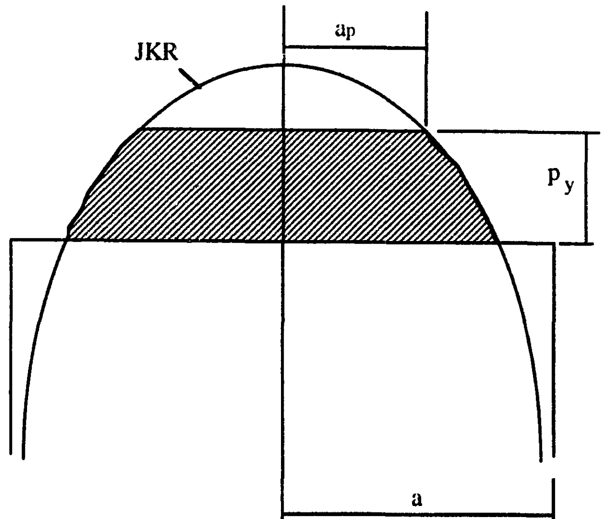

Fig. 7. Normal traction distribution for adhesive, elastic-perfectly plastic spheres.

$$a=\left(\frac{3R^*P_1}{4E^*}\right)^{1/3} \tag{59}$$

where

$$P_1=P+2P_c\pm(4PP_c+4P_c^2)^{1/2} \tag{60}$$

is the effective Hertzian force which would produce the same contact area, P is the applied force and P_c is the pull-off force defined by Eq. (50). The relative approach and contact force may also be expressed [3] as

$$\alpha=\frac{a^2}{R^*}-\left(\frac{2\pi\Gamma a}{E^*}\right)^{1/2} \tag{61}$$

$$P=\frac{4E^*a^3}{3R^*}-(8\pi\Gamma E^*a^3)^{1/2} \tag{62}$$

In order to model elastic–perfectly plastic spheres with adhesion, we again assume a limiting contact pressure p_y, as shown in Fig. 7. The applied force during plastic deformation, P_p, is obtained from

$$P_p=P-2\pi\int_0^{a_p}[p(r)-p_y]r\,\mathrm{d}r \tag{63}$$

which leads to

$$P_p=P-\frac{4E^*a^3}{3R^*}\left[1-\left(1-\frac{a_p^2}{a^2}\right)^{3/2}\right]+(8\pi\Gamma E^*a^3)^{1/2}\left[1-\left(1-\frac{a_p^2}{a^2}\right)^{1/2}\right]+\pi a_p^2p_y \tag{64}$$

with P defined by Eq. (62). Using Eq. (58), the limiting contact pressure is given by

$$p_y=\frac{2E^*a_y}{\pi R^*}-\left(\frac{2\Gamma E^*}{\pi a_y}\right)^{1/2} \tag{65}$$

Unfortunately, for the case of adhesive spheres, Eq. (17) is not exactly correct but we nevertheless assume it to be a reasonable approximation in order to simplify Eq. (64) and obtain

$$P_p=\frac{4E^*a_y^3}{3R^*}-a_y(8\pi\Gamma E^*a)^{1/2}+\pi p_y(a^2-a_y^2) \tag{66}$$

Differentiating Eqs. (66) and (61),

$$\frac{dP_p}{da}=2\pi p_y a-\left(\frac{2\pi\Gamma E^*a_y^2}{a}\right)^{1/2} \tag{67}$$

$$\frac{d\alpha}{da}=\frac{2a}{R^*}-\left(\frac{\pi\Gamma}{2E^*a}\right)^{1/2} \tag{68}$$

By combining Eqs. (67) and (68) and using suitable substitutions it can be shown that the contact stiffness during plastic deformation is given by

$$\frac{dP_p}{d\alpha}=\frac{3\pi R^*p_y\sqrt{P_1}-2E^*a_y\sqrt{P_c}}{3\sqrt{P_1}-\sqrt{P_c}} \tag{69}$$

During elastic loading the contact stiffness was given by Thornton and Yin [12] as

$$\frac{dP}{d\alpha}=2E^*a\left[\frac{3-3\left(\frac{a_c^3}{a^3}\right)^{1/2}}{3-\left(\frac{a_c^3}{a^3}\right)^{1/2}}\right] \tag{70}$$

where

$$a_c^3=\frac{3R^*P_c}{4E^*} \tag{71}$$

Substituting Eqs. (59) and (71) into Eq. (70) gives

$$\frac{dP}{d\alpha}=2E^*a\left[\frac{3\sqrt{P_1}-3\sqrt{P_c}}{3\sqrt{P_1}-\sqrt{P_c}}\right] \tag{72}$$

During elastic recovery, it was shown by Ning [13] that the contact stiffness is provided by

$$\frac{dP}{d\alpha}=2E^*a\left[\frac{3\sqrt{P_{1r}}-3\sqrt{P_{cr}}}{3\sqrt{P_{1r}}-\sqrt{P_{cr}}}\right] \tag{73}$$

where

$$a^3=\frac{3R_p^*R_{1r}}{4E^*} \tag{74}$$

$$P_{1r}=P+2P_{cr}\pm\sqrt{4PP_{cr}+4P_{cr}^2} \tag{75}$$

$$R_p^*=\frac{R^*P_1^*}{P^*+\sqrt{4P_cP_1^*}} \tag{76}$$

and

$$P_{cr}=\frac{3}{2}\pi\Gamma R_p^* \tag{77}$$

where P_1^* is the value of the equivalent Hertzian force, given by Eq. (60), from which unloading commenced. Fig. 8 illustrates the form of the force–displacement relationship for elastic–plastic loading and elastic unloading, as obtained from numerical simulations based on the above theoretical model. It can be seen that, not only does the unloading stiffness increase with increase in impact velocity but also, the pull-off force required to overcome the adhesion increases with impact velocity due to the locally increased radius of curvature, as indicated by Eq. (77).

By combining the above theory with Newton's laws of motion, Ning [13] solved the impact problem numerically. Fig. 9 shows the effect of interface energy on the coefficient of restitution. For the case illustrated, the yield velocity is 0.62 m s^{-1}. It can be seen that the coefficient of restitution is sensitive to the value of interface energy at velocities below the yield velocity but is not significantly dependent on the interface energy when the impact velocity is greater than the yield velocity. The effect of the limiting contact pressure (and hence yield velocity) on the coefficient of restitution is shown in Fig. 10 for an interface energy of 0.2 J m^{-2}. When the limiting contact pressure $p_y=3.04$ GPa the yield velocity $V_y=0.62$ m s^{-1} is more than an order of magnitude greater than the sticking velocity $V_s=0.016$ m s^{-1} and the maximum value of the coefficient of restitution $e\approx1.0$. When the limiting contact pressure is reduced to $p_y=1.0$ GPa, the sticking velocity is not affected but the yield velocity is reduced to $V_y=0.045$ m s^{-1} and the maximum value of $e\approx0.9$. If the limiting contact pressure is reduced to $p_y=0.5$ GPa then plastic deformation occurs at zero load, i.e., $V_y=0$, and the maximum value of $e\approx0.5$. For this case, as shown in Fig. 10, the fact that the yield velocity is less than the sticking velocity predicted for elastic–adhesive spheres by Eq. (54) results in a higher sticking velocity: $V_s=0.032$ m s^{-1}. In all cases, if

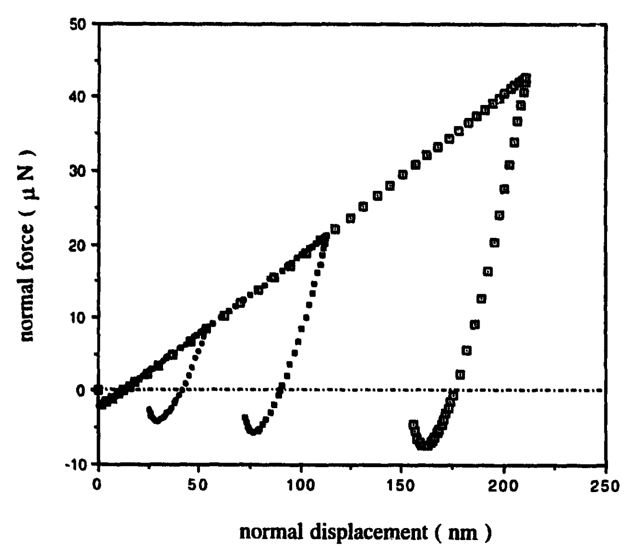

Fig. 8. Effect of impact velocity on the normal contact force–displacement response for adhesive, elastic–perfectly plastic spheres.

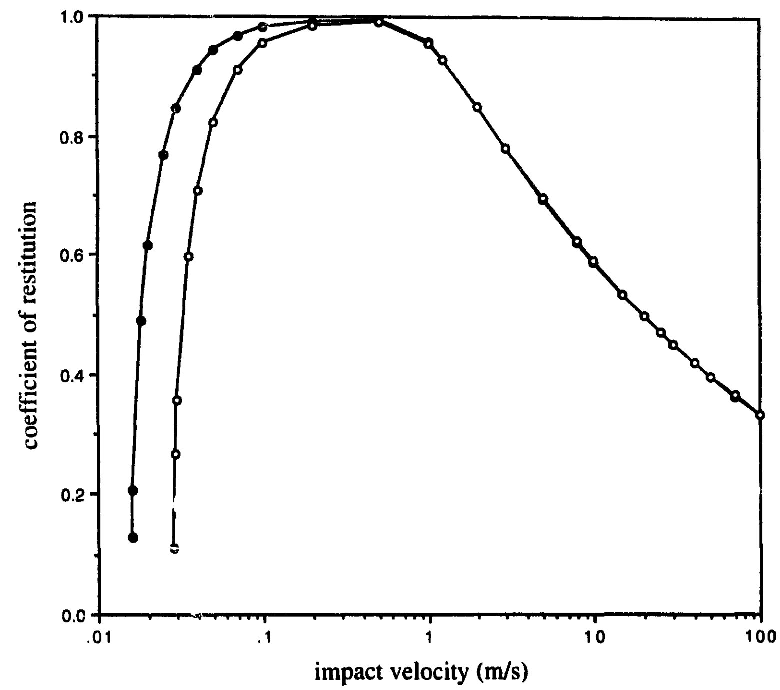

Fig. 9. Effect of interface energy on the coefficient of restitution (●: $\Gamma=0.2\ \mathrm{J\ m^{-2}}$, ○: $\Gamma=0.4\ \mathrm{J\ m^{-2}}$).

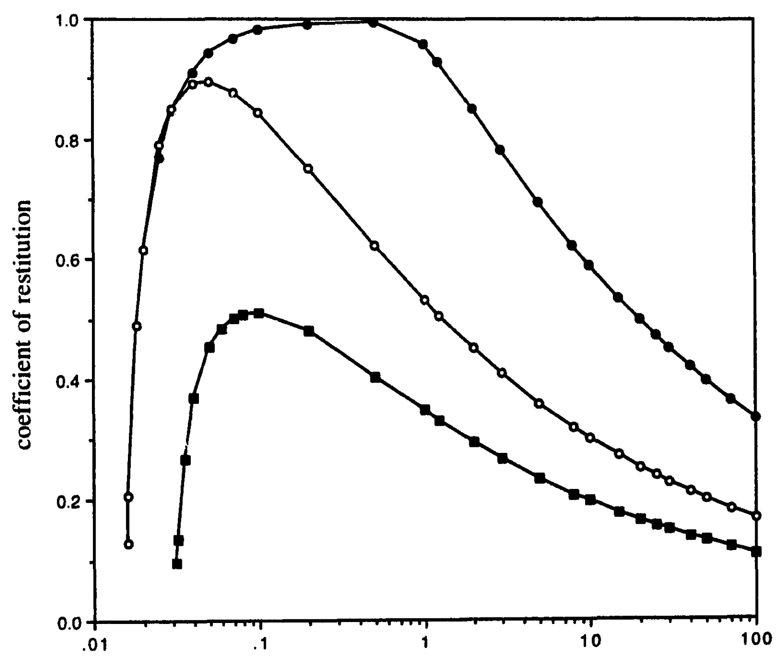

Fig. 10. Effect of limiting contact pressure on the coefficient of restitution (●: $p_y=3.04$ GPa, ○: $p_y=1.0$ GPa, ■: $p_y=0.5$ GPa).

the impact velocity $V_i>10V_y$ the results follow a power law relationship with an exponent of $-1/4$.

Due to the complexity of the equations which define the behaviour of adhesive elastic–perfectly plastic spheres it has not been possible to obtain an analytical solution in the way that this was obtained for non-adhesive spheres. However, an analytical solution can be obtained by assuming that the work dissipated due to plastic deformation and the work dissipated due to adhesive rupture are additive. That is to say

$$(1-e^2)=(1-e_p^2)+(1-e_a^2) \tag{78}$$

where e_p is the coefficient of restitution due to plastic deformation given by Eq. (44) and e_a is the coefficient of restitu-

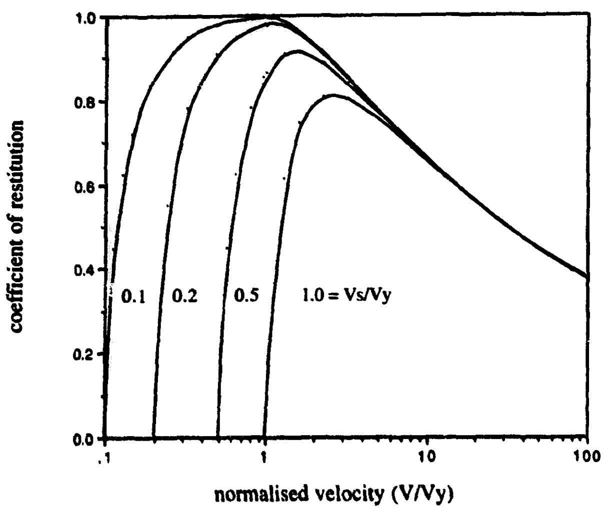

Fig. 11. Theoretical prediction of the velocity dependent coefficient of friction.

tion due to adhesive rupture given by Eq. (57). This leads to the following equations:

for $V_i \leq V_s$

$$e=0 \tag{79}$$

for $V_s \leq V_i \leq V_y$

$$e=\left[1-\left(\frac{V_s}{V_i}\right)^2\right]^{1/2} \tag{80}$$

and for $V_i \geq V_y$

$$e^2= \frac{6\sqrt{3}}{6}\left[1-\frac{1}{6}\left(\frac{V_y}{V_i}\right)^2\right] \times \left[\frac{\left(\frac{V_y}{V_i}\right)}{\left(\frac{V_y}{V_i}\right)+2\sqrt{\frac{6}{5}-\frac{1}{5}\left(\frac{V_y}{V_i}\right)^2}}\right]^{1/2} -\left(\frac{V_s}{V_i}\right)^2 \tag{81}$$

The smooth curves superimposed on Fig. 10 were obtained using the above equations and, as can be seen, agree well with the numerically simulated results obtained by Ning [13]. Finally, the solution to the above equations is illustrated in Fig. 11 by plotting the coefficient of restitution against the normalised velocity (V_i/V_y) for different ratios of (V_s/V_i).

6. Conclusions

An analytical solution for the coefficient of restitution for normal impact of adhesive, elastic–perfectly plastic spheres has been obtained. The solution is expressed in terms of the impact velocity, the critical sticking velocity and the velocity below which the interaction is assumed to be elastic. If the impact velocity is less than or equal to the sticking velocity the particles remain adhered to each other and $e=0$. If the impact velocity is greater than the sticking velocity then, as the impact velocity increases, the coefficient of restitution increases at a decreasing rate to a maximum value and then decreases at an increasing rate until, at impact velocities more than ten times the yield velocity, the coefficient of restitution is a function of the impact velocity raised to the power $-1/4$. The maximum value of the coefficient of restitution decreases with increase in the ratio of the sticking velocity to the yield velocity.

The analytical solution provided by Eqs. (79)–(81) can easily be incorporated into numerical simulations of gas-particle flows to provide variable coefficients of restitution which are velocity dependent and hence will lead to more realistic distributions of energy dissipation in complex flow fields. The equations are expressed in terms of parameters which may be determined by simple impact experiments rather than in terms of material properties such as the yield stress and interface energy which are difficult to reliably ascertain for many materials.

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